Classification Assignment

Chronic Kidney Disease (CKD) Prediction Report

Problem Statement

The objective of this project is to assist Hospital Management by developing a predictive model that can identify **Chronic Kidney Disease (CKD)** in patients based on various clinical parameters. This is a binary classification problem where the goal is to accurately predict whether a patient has the disease (1) or not (0).

1.Identify your Problem statement

The problem statement for this project is to fulfill a requirement from Hospital Management to create a predictive model. This model will predict Chronic Kidney Disease (CKD) based on several clinical parameters provided in the dataset.

2. Dataset Information

Total number of rows: **399**

Total number of columns: **28**

Target variable: **classification**

3. Data Preprocessing

The following preprocessing steps were applied:

- Missing values handled using mean/mode
- Categorical data converted into numerical using Label Encoding
- Feature scaling applied using StandardScaler (for KNN, SVM)
- Dataset split into training and testing sets

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Feature Type	Original Data (Example)	Pre-processed Data
Nominal (Binary)	**"normal", "abnormal"**	**1, 0**
Nominal (Multi-class)	**"a", "b", "c", "d", "e"**	**Multi-column binary flags (One-Hot Encoding)**
Target Variable	**"yes" (CKD), "no" (Not CKD)**	**1, 0**
Numeric	**76.45, 148.11**	**Kept as Float/Int**

4.Model Building & Evaluation

Algorithm	Best Use Case	Complexity	Accuracy	F1-Score (Weighted)	Status
SVM (Grid)	High-dimensional data; excellent for complex clinical parameters.	High	**0.99**	**0.99**	**Top Performer**
Logistic Regression	Linear relationships; provides a fast and reliable baseline.	Low	**0.99**	**0.99**	**Reliable Baseline**
Random Forest (Grid)	Reducing variance; handles noisy data and outliers well.	High	0.98	0.98	Strong & Robust
Naive Bayes	Large datasets; extremely fast and efficient processing.	Low	0.98	0.98	Highly Efficient
Decision Tree (Grid)	Highly interpretable "if-then" logic for medical decisions.	Medium	0.93	0.93	Interpretability
KNN (Grid)	Finding similar patient profiles based on "neighboring" data.	Medium	0.79	0.79	Fair

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1.SVM (Grid)

```
print("The report:\n",clf_report) #SVM
```

```
The report:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        51
           1       1.00      0.99      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

2.Decision Tree (Grid)

```
print("The report:\n",clf_report) #Decision Tree
```

```
The report:
              precision    recall  f1-score   support

           0       0.92      0.90      0.91        51
           1       0.94      0.95      0.95        82

    accuracy                           0.93       133
   macro avg       0.93      0.93      0.93       133
weighted avg       0.93      0.93      0.93       133
```

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3. Random Forest (Grid)

```
print("The report:\n",clf_report) #Random Forest
```

```
The report:
               precision    recall  f1-score   support

           0       0.98      0.98      0.98        51
           1       0.99      0.99      0.99        82

    accuracy                           0.98       133
   macro avg       0.98      0.98      0.98       133
weighted avg       0.98      0.98      0.98       133
```

4. KNN (Grid)

```
print("The report:\n",clf_report) #Grid KNN
```

```
The report:
               precision    recall  f1-score   support

           0       0.67      0.90      0.77        51
           1       0.92      0.72      0.81        82

    accuracy                           0.79       133
   macro avg       0.79      0.81      0.79       133
weighted avg       0.82      0.79      0.79       133
```

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5.Logistic Regression Grid Model

```
print("The report:\n",clf_report) #Logistic Regression
```

```
The report:
               precision    recall  f1-score   support

           0       0.98      1.00      0.99        51
           1       1.00      0.99      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

6.Navie Baye's

```
print("The report:\n",clf_report) #Grid Navie bayes
```

```
The report:
               precision    recall  f1-score   support

           0       0.94      1.00      0.97        51
           1       1.00      0.96      0.98        82

    accuracy                           0.98       133
   macro avg       0.97      0.98      0.98       133
weighted avg       0.98      0.98      0.98       133
```

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6.Final Model

Rank	Model	Macro F1-score	Status
1	SVM	0.99	Final Model
2	Logistic Regression	0.99	Excelent
3	Random Forest	0.98	High
4	Navie Bayes	0.98	High
5	Decision Tree	0.93	Medium
6	KNN	0.79	Poor